Eunsung Jang

Integrated M.S./Ph.D. student, RF System Technologies Lab., Department of Electrical and Computer Engineering, Ajou University, Suwon, Republic of Korea
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RESEARCH INTERESTS

- Dual-polarized distributed FMCW radar with lens array antennas for through-the-wall human detection
- AI radar signal processing for through-the-wall human detection
- Wide-angle scanning phased array antenna systems for anti-drone detection
- Low-loss coaxial cables

EDUCATION

Ajou University, Suwon, Korea [CONFIRM: start year]-present
Integrated M.S./Ph.D. program, Department of Electrical and Computer Engineering. Advisor: Prof. Jeong-Wook Kim

[CONFIRM: university] [CONFIRM: year]
B.S. in [CONFIRM: major]

PROFESSIONAL EXPERIENCE

Hanwha Ocean, Shipbuilding Division 2015-2017
HVAC Design Engineer

PUBLICATIONS

Journal articles

[J1] **E. Jang**, J.-W. Park, K.-H. Park, J.-Y. Lee, H.-S. Kim, and J.-W. Kim, "Design of dual-polarized distributed FMCW radar system with high-gain lens array antenna for through-the-wall human detection application," IEEE Internet of Things Journal, 2025.

[J2] **E. Jang**, E. You, and J.-W. Kim, "Coaxial cable loss reduction through uniform air-hole PTFE dielectric fabricated by ram extrusion," under review.

International conference

[C1] **E. Jang**, J.-W. Park, K. Park, H. Kim, J. Lee, and J.-W. Kim, "Machine learning selection of range bins for coherent integration in distributed radar observing frontal and lateral motion," in Proc. Asia-Pacific Microwave Conference (APMC), 2025.

Domestic conferences (in Korean)

[D1] **E. Jang**, S. Choi, M. Cho, J.-W. Park, S. Kim, and J.-W. Kim, "Design of a dual-polarized high-gain lens-coupled stacked patch antenna unit cell for C-band ISAC applications," KICS Summer Conference, 2026.

[D2] **E. Jang**, J.-W. Park, K.-H. Park, J.-Y. Lee, H.-S. Kim, and J.-W. Kim, "Through-the-wall human detection using band-energy-difference-based channel combining in a distributed dual-polarized FMCW radar," KIEES Winter Conference, 2026.

[D3] **E. Jang**, J.-W. Park, J.-W. Kim, K.-H. Park, J.-Y. Lee, and H.-S. Kim, "Dual-polarized distributed FMCW radar for through-the-wall human detection," KIEES Summer Conference, 2025.

AWARDS AND SCHOLARSHIPS

2026 Master's Excellence Scholarship (Science and Engineering), [CONFIRM: awarding body]

2025 Gold Prize, BK21 Graduate School Innovation Outstanding Paper Award [CONFIRM: 2025 or 2026]

2025 Encouragement Award, 24th Graduate Student Paper Contest in Radio Engineering, for "Design of a dual-polarized distributed FMCW radar system with a high-gain lens array antenna and its application to through-the-wall human detection"

2024 Encouragement Award, Industry-Academia Fair, Department of Electrical and Computer Engineering, Ajou University

RESEARCH EXPERIENCE

- **Through-the-wall human detection radar.** Designed a dual-polarized distributed FMCW radar with high-gain lens array antennas; developed machine learning range-bin selection and polarization channel combining [J1, C1, D2, D3].
- **Wide-angle dual-polarized phased array for drone detection.** X-band, ±80° beam scanning, LHCP/RHCP. Carried out design, fabrication, measurement, test-jig design, chamber measurement at RAPA, and beamforming experiments on my own.
- **C-band ISAC antenna.** Dual-polarized high-gain lens-coupled stacked patch antenna unit cell [D1].
- **Low-loss coaxial cable.** EM simulation, design optimization, manuscript writing, and response to reviewers [J2].
- **X-band analog beamforming SAR.** Wrote the ADAR1000 SPI beam-control manual and a debugging protocol for reverse voltage in the receiver.
- **Antenna design automation.** Built a workflow in which a large language model drives a 3D EM simulator from text instructions; modeling time fell by more than 95%, and the loop ran over 1,000 simulations in 5 days.
- **Research proposals.** Designed the structure of and coordinated five national R&D proposals, each written in 1-2 days.

SKILLS

- **EM design:** 3D full-wave EM simulation, Keysight ADS, openEMS, MATLAB
- **RF measurement:** VNA, spectrum analyzer, oscilloscope, signal generator, power meter, anechoic-chamber antenna measurement
- **Hardware:** PCB design (PADS), soldering, breadboard prototyping, test-jig design, ADAR1000 control over SPI
- **Programming:** Python, MATLAB, JavaScript, Java (Android), VBA
- **Languages:** Korean (native), English [CONFIRM: test score, if any]